

Computational Logic

2- Proofs in Propositional Logic

(lab classes 3, 4 and 5)

All deductions must use the DN natural deduction system and the Fitch notation given in class.

Notes for obtaining a deduction from a formula φ

- If the formula φ is an implication $\psi \rightarrow \theta$, assume ψ and deduce θ; then apply the rule of introduction of implication.
- If the formula φ is a negation $\neg\psi$, assume ψ and deduce F; then apply the rule of introduction of negation.
- By contradiction: assume $\neg\varphi$ and deduct F. In case there is a negation $\neg\psi$ in the premises, one can deduce ψ to get F.
- Suppose one of the premises is $\varphi \vee \psi$ and one wants to deduce γ. We can: assume φ and deduce γ, assume ψ and deduce γ; then apply the rule of elimination of the disjunction.
- Suppose an assumption is $\neg\varphi$ and we want to get **F**: we can try to get φ and apply the intro rule of **F**.
- Suppose we have an implication $\phi \rightarrow \theta$. We can then try to deduct ϕ and apply the rule of elimination of implication (modus ponens) to get θ.

Note that each step of the deduction must be obtained by applying a rule or be a repetition of a formula already deduced or a premise. Thus each deduction step must either be an assumption or result from the application of a rule to formulas already deduced, so "semantic equivalences" cannot be used.

Note: In case you use derived rules, you should prove then separately.

2.1. Some of the following arguments are valid and some are not. Use symbols to write the logical form of each one. If the argument is valid, identify the inference rule.

(a) If Hugo solves the exercises correctly, he will travel to the Azores. Hugo is going to travel to the Azores. So, Hugo correctly solved the exercises.

(b) If I go to the movies, I don't solve the exercises. If I don't solve the exercises I will fail math. So if I go to the movies I'm going to fail math.

2.2. Use the inference rules studied in lectures to complete the following arguments so that they are valid.

(a) If math is hard, then Santa Claus exists. Santa Claus doesn't exists. Therefore...

(b) If math is hard, then Santa Claus exists. Math is hard. Therefore...

⇒ **2.3** Match each of the following tautologies to one of the logical arguments.

Tautologies

(1) $p \vee \neg p$

(2) $p \wedge q \rightarrow p$

(3) $p \rightarrow p \vee q$

(4) $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$

(5) $((p \rightarrow q) \wedge (\neg p \rightarrow q)) \rightarrow q$

(6) $(((p \vee q) \rightarrow r) \wedge p) \rightarrow r$

Arguments

(a) John has enrolled in Computational Logic. Then John is taking Computational Logic or he is taking Data Structures and Algorithms.

(b) John can program in C and Java. Therefore, John knows how to program in Java.

(c) John gets frequency if he does his work in C or Java. John did the work in C. So John obtained frequency.

(d) Mary might have gone to the Computational Logic class today or not.

(e) If it rains, I will go to the movies. Then, if I did not go to the movies, it did not rain.

(f) If it snows I am staying at home; if it doesn't snow I am still staying at home. Therefore I am staying at home.

⇒ **2.4** Indicate whether the following deductions are valid. If so, indicate the rules used in each step. If not valid, indicate which step it is not correct and if there is a correct deduction with the same premises

and conclusion.

a)

1	$p \vee q$
2	| p
3	| p
4	| q
5	| q
6	$p \wedge q$

b)

1	$(p \wedge q) \vee r$
2	| $p \wedge q$
3	| p
4	| $p \vee r$
5	| r
6	| $p \vee r$
7	$p \wedge r$

c)

1	$\neg(p \vee \neg q)$
2	| $\neg q$
3	| $p \vee \neg q$
4	| $\mathbf{F}$
5	$\neg\neg q$
6	q

d)

1	$p \to q$
2	| p
3	| q
4	| $\neg p \vee q$
5	$\neg p \vee q$

e)

1	$p \to \neg p$
2	| p
3	| $\neg p$
4	| $\mathbf{F}$
5	$\neg p$

f)

1	$\neg p \to \neg q$
2	| q
3	| $\neg q$
4	| $\mathbf{F}$
5	| p
6	$q \to p$

⇒ **2.5** Show that:

(a) $(p \wedge q) \to r, p \vdash q \to r$
(b) $q \to p, q \to \neg p \vdash \neg q$
(c) $(\neg p \vee \neg q) \vee r, q \vee r, p \vdash r$
(d) $p \vee \neg q, \neg r \to q, r \to \neg s, s \vdash p$
(e) $p \to (q \vee r), q \to s, r \to t \vdash p \to (s \vee t)$.
(f) $p \to (q \wedge r), q \vee s \to t, p \vee s \vdash t$.
(g) $(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k) \vdash \neg r$.

⇒ **2.6** Show that:

(a) $(\delta \wedge \neg\varphi) \to \psi, \neg\psi, \delta \vdash \varphi$
(b) $\vdash (\delta \to \psi) \to ((\neg\delta \to \neg\varphi) \to (\varphi \to \psi))$
(c) $\vdash (\delta \wedge \varphi) \to \varphi$
(d) $\vdash \delta \to (\neg\varphi \to \neg(\delta \to \varphi))$
(e) $\vdash ((\varphi \to \delta) \wedge (\delta \to \varphi)) \to ((\varphi \vee \delta) \to (\varphi \wedge \delta))$

⇒ **2.7** Using rules of logical inference, build a proof for the conclusion "It rained" from the hypothesis "If it doesn't rain or if there is no fog, then the sailing will take place and the rescue demonstration will be carried out", "If the sailing competition takes place then the trophy will be awarded" and "The trophy was not awarded".

2.8. Show that:

(a) $\delta \to (\varphi \to \psi), \neg\psi, \delta \vdash \neg\varphi$

(b) $\vdash (\varphi \vee \psi) \to (\psi \vee \varphi)$

(c) $\vdash (\delta \to (\varphi \to \psi)) \to (\varphi \to (\delta \to \psi))$

(d) $\vdash \delta \to (\delta \vee \varphi)$

(e) $\vdash ((\varphi \to \delta) \to \delta) \to ((\delta \to \varphi) \to \varphi)$

(f) $\delta, \varphi \to \psi \vdash \varphi \to (\delta \to \psi)$

(g) $\delta, \neg\varphi \vdash \neg(\delta \to \varphi)$

$\Rightarrow$ **2.9** Indicate which of the following deductions are valid and which are not:

(a) $\neg\delta, \delta \vee \varphi \vdash \varphi$

(b) $\varphi \wedge \neg\varphi \vdash \neg(\delta \to \psi) \wedge (\delta \to \psi)$

(c) $\neg(\neg\delta \vee \varphi) \vdash \delta$

(d) $\delta \vee \psi, \neg\psi \vee \varphi \vdash \delta \vee \varphi$

(e) $\vdash (\neg\delta \to \delta) \to \delta$

(f) $\neg\delta \to \neg\psi \vdash \neg\psi \to \neg\delta$

$\Rightarrow$ **2.10** Justify the validity or falsity of each of the following statements with a proof or a counter-example:

(a) if $\vdash \psi$, then $\delta_1, \ldots, \delta_n \vdash \psi$;

(b) if $\delta_1, \ldots, \delta_n \vdash \psi$, then $\vdash \psi$;

(c) if $\delta_1, \delta_2 \vdash \psi$, then $\delta_2, \delta_1 \vdash \psi$;

(d) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1, \delta_2 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;

(e) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;

(f) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then $\delta_1 \vdash \psi$;

(g) if $\delta_1 \vdash \varphi \to \psi$ and $\delta_2 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;

(h) $\delta, \delta \vdash \psi$ if and only if $\delta \vdash \psi$.

$\Rightarrow$ **2.11** Show, without using the completeness of DN, the following deductive equivalences. Note that you have to show that the second formula is deduced from the first formula and vice versa.

(a) $\neg\delta_1 \wedge \delta_2 \vdash \delta_1 \vee \delta_2$

(b) $\delta_1 \wedge \delta_2 \vdash \delta_1 \vee \delta_2$

(c) $(\psi \wedge \varphi) \wedge \delta \dashv\vdash \psi \wedge (\varphi \wedge \delta)$

(d) $\neg(\neg\delta_1 \wedge \neg\delta_2) \dashv\vdash \delta_1 \vee \delta_2$

(e) $\delta_1 \to \delta_2 \dashv\vdash \neg(\delta_1 \wedge \neg\delta_2)$

(f) $\delta_1 \wedge \delta_2 \dashv\vdash \neg(\neg\delta_1 \vee \neg\delta_2)$

(g) $(\varphi \vee \psi) \to \delta \dashv\vdash (\varphi \to \delta) \wedge (\psi \to \delta)$

(h) $(\delta \vee \varphi) \wedge (\delta \vee \psi) \dashv\vdash \delta \vee (\varphi \wedge \psi)$

(i) $\varphi \to (\psi \wedge \delta) \dashv\vdash (\varphi \to \psi) \wedge (\varphi \to \delta)$

(j) $\delta_1 \wedge \neg\delta_2 \vdash \neg(\delta_1 \to \delta_2)$

2.12. Show, without using the completeness of DN, the following deductive equivalences. Note that you have to show that the second formula is deduced from the first formula and vice versa.

(a) $\neg(\delta_1 \wedge \delta_2) \dashv\vdash \neg\delta_1 \vee \neg\delta_2$

(b) $\neg(\delta_1 \vee \delta_2) \dashv\vdash \neg\delta_1 \wedge \neg\delta_2$

(c) $\delta_1 \to \delta_2 \dashv\vdash \neg\delta_1 \vee \delta_2$

(d) $\neg(\delta_1 \to \delta_2) \dashv\vdash \delta_1 \wedge \neg\delta_2$

(e) $\neg\delta_1 \wedge \neg\delta_2 \vdash \delta_1 \to \delta_2$

(f) $\delta_1 \wedge \neg\delta_2 \vdash \neg(\delta_1 \wedge \delta_2)$

(g) $(\psi \vee \varphi) \vee \delta \dashv\vdash \psi \vee (\varphi \vee \delta)$

(h) $\delta \to \varphi \dashv\vdash (\delta \to (\delta \wedge \varphi)) \wedge ((\delta \wedge \varphi) \to \delta)$

(i) $\delta \to \varphi \dashv\vdash (\varphi \to (\delta \vee \varphi)) \wedge ((\delta \vee \varphi) \to \varphi)$

(j) $(\delta \to \varphi) \wedge (\varphi \to \delta) \dashv\vdash (\delta \vee \varphi) \to (\delta \wedge \varphi)$

(k) $(\delta \to \varphi) \dashv\vdash (\neg\varphi \to \neg\delta)$

(l) $\varphi \wedge (\delta \vee \psi) \dashv\vdash (\varphi \wedge \delta) \vee (\varphi \wedge \psi)$

2.13. Show, without using the completeness of the natural deduction, that for every set of formulas in propositional logic Σ and formula δ, if you have:

(a) If $\Sigma \vdash \neg\delta$ then $\Sigma \cup \{\delta\}$ is inconsistent.

(b) $\Sigma \models \delta$ if and only if $\Sigma \cup \{\neg\delta\}$ is unsatisfiable.

(c) $\Sigma \cup \{\neg\delta\} \vdash \mathbf{F}$ if and only if $\Sigma \vdash \delta$

2.14. Justify the truth or falsity of the following statements. If the statement is false, you can provide a counter-example.

(a) Let Σ be a set of propositional logic formulas, then the following are equivalent:

1. Σ is inconsistent.
2. $\Sigma \vdash \delta$ whatever the propositional logic formula δ.

(b) Let Σ and Γ be two sets of propositional logic formulas, then the following are equivalent:

1. $\Sigma \cup \Gamma$ is satisfiable.
2. $\Sigma \cup \{\delta\}$ is satisfiable whatever the formula $\delta \in \Gamma$.

(c) Let Σ and Γ be two sets of formulas in propositional logic. Then

$$\{\delta \mid \Sigma \cup \Gamma \vdash \delta\} = \{\delta \mid \Sigma \vdash \delta\} \cup \{\delta \mid \Gamma \vdash \delta\}.$$

⇒ **2.15** Let Σ and Γ be sets of propositional logic formulas such that for any formula δ one has $\Sigma \vdash \delta$ or $\Gamma \vdash \delta$.

(a) Show that one of the sets Σ or Γ is not satisfiable.

(b) Show that $\Sigma \cup \Gamma$ is not satisfiable.

2.16. Let Σ and Γ be sets of formulas from the propositional logic, such that for every formula δ we have:

$$\Sigma \vdash \delta \text{ if and only if } \Gamma \vdash \delta.$$

Justify the validity or falsity of each of the following statements with proofs or counter-examples:

(a) Let $\delta_1, \ldots, \delta_n$ be a deduction of a formula δ from the set Σ. Then $\delta_1, \ldots, \delta_n$ is also a deduction from δ from Γ.

(b) Let $\overline{\Gamma} = \{\neg\delta \mid \delta \in \Gamma\}$ and $\delta_1, \ldots, \delta_n$ be a deduction of a formula δ from the set Γ, then $\neg\delta_1, \ldots, \neg\delta_n$ is a deduction from $\neg\delta$ from the set of hypotheses $\overline{\Gamma}$;

(c) Σ is satisfiable if and only if Γ is satisfiable.

Other deductive systems: *tableau* and resolution

⇒ **2.17** Construct deductions of semantic *tableau* for the following formulas. A deduction of a formula φ by *tableau* is a refutation, that is, from $\neg\varphi$ one tries to deduce **F**.

(a) $\neg((p \vee q) \wedge (\neg p \wedge \neg q))$

(b) $((p \to q) \wedge (q \to r)) \to (p \to r)$

(c) $(\neg p \to q) \to ((p \to q) \to q)$

(d) $((p \to q) \to p) \to p$

2.18. Consider the following formula in conjunctive normal form (FNC):

$$(p \vee \neg q \vee r) \wedge \neg p \wedge (q \vee r \vee p) \wedge (p \vee \neg r)$$

(a) Convert the formula to a set of clauses.

(b) From this set, construct a deduction of **F**, using only the resolution rule.

⇒ **2.19** Find a deduction of **F** (a refutation) for:

$$\{\{\neg p, \neg q, r\}, \{p, r\}, \{q, r\}, \{\neg r\}\}$$

2.20. Using the resolution rule, find a deduction by resolution of:

$$((p \to q) \wedge (q \to r)) \to \neg(\neg r \wedge p)$$

⇒ **2.21** Construct deductions using *tableau* and **resolution** for the following formulas:

(a) $(p \to q) \to ((\neg p \to q) \to q)$

(b) $q \to (p \to (q \wedge p))$

(c) $q \to (\neg p \to \neg(q \to p))$

2.22. Construct deductions from *tableau* and by **resolution** for the items of exercise 2.8 that do not have premises.

Tableau

α	α_1	α_2	β	β_1	β_2
$\varphi \wedge \psi$	φ	ψ	$\neg(\varphi \wedge \psi)$	$\neg\varphi$	$\neg\psi$
$\neg(\varphi \vee \psi)$	$\neg\psi$	$\neg\varphi$	$\varphi \vee \psi$	φ	ψ
$\neg(\varphi \rightarrow \psi)$	φ	$\neg\psi$	$\varphi \rightarrow \psi$	$\neg\varphi$	ψ

Table 1: Uniform notation: formulas α and β

$\frac{\neg\neg\varphi}{\varphi}$	$\frac{\neg\mathbf{F}}{\mathbf{T}}$	$\frac{\neg\mathbf{T}}{\mathbf{F}}$	$\frac{\alpha}{\alpha_1 \quad \alpha_2}$	$\frac{\beta}{\beta_1 \mid \beta_2}$

Table 2: Expansion rules from *tableaux*

Resolution

Let C, C' be clauses, the resolution rule is

$$\frac{C \cup \{p\} \qquad C' \cup \{\neg p\}}{C \cup C'}$$

A deduction of $\mathbf{F}$ from a set C of clauses is called a refutation of C.

Natural Deduction (DN) for propositional logic (Fitch notation)

	Introduction	Elimination
$\wedge$	φ $\vdots$ ψ $\vdots$ $\varphi \wedge \psi$ $\wedge$I	$\vdots$ $\varphi \wedge \psi$ $\vdots$ φ $\wedge$E $\vdots$ $\varphi \wedge \psi$ $\vdots$ ψ $\wedge$E
$\vee$	$\vdots$ φ $\vdots$ $\varphi \vee \psi$ $\vee$I $\vdots$ φ $\vdots$ $\psi \vee \varphi$ $\vee$I	$\varphi \vee \psi$ $\vdots$ φ / $\vdots$ / γ ψ / $\vdots$ / γ γ $\vee$E
$\neg$	φ / $\vdots$ / **F** $\neg\varphi$ $\neg$I	$\vdots$ $\neg\neg\varphi$ $\vdots$ φ $\neg$E
F	φ $\vdots$ $\neg\varphi$ $\vdots$ **F FI**	$\vdots$ **F** $\vdots$ φ **FE**
$\rightarrow$	φ / $\vdots$ / ψ $\varphi \rightarrow \psi$ $\rightarrow$I	φ $\vdots$ $\varphi \rightarrow \psi$ $\vdots$ ψ $\rightarrow$E